Varunprasaath R. Selvaraju, Ph.D.

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PROFESSIONAL SUMMARY

PhD chemist/materials scientist with 6+ years in analytical characterization, structure-property/stability analysis, and DOE-guided process development. Applies LC-MS, NMR/GPC, FT-IR, UV-Vis-NIR, TGA/DSC, QCM-D, and electrochemistry to build reproducible process windows and technical documentation.

CORE COMPETENCIES

Analytical Method Development | Small-Molecule Characterization | Structure Elucidation | Solid-State/Thermal Analysis | Stability Analysis | CMC Documentation

WORK EXPERIENCE

University of Colorado Boulder
July 2025 - Present

Postdoctoral Researcher

- Lead DOE-guided bio-based polymer process development, converting material-quality data into reproducible process windows and scale-up specifications.

University of Colorado Boulder
July 2021 - July 2025

Graduate Researcher

- Designed and characterized organic molecular materials, mapping structure, purity, and stability across 15+ variants with spectroscopy, NMR/GPC, thermal analysis, electrochemistry, and MS.

Georgia Institute of Technology
Aug 2019 - July 2021

Graduate Researcher

- Drove AFOSR-funded thin-film coating process development, linking CVD/surface preparation to morphology, interface quality, and optical performance.

ETH Zurich
May 2017 - July 2017

Research Assistant

- Optimized CVD polymer thin-film deposition with DOE and in-situ QCM-D monitoring to define growth kinetics and reproducible process controls.

University of Colorado Boulder
2021 - 2025

Laboratory Operations Manager

- Managed characterization/process-development lab operations for a 15-member group; implemented calibration, SOP, and safety systems that improved instrument uptime by 30%.

Georgia Tech
2019-2021

Teaching Assistant, Organic Chemistry I & Lab

- Mentored 120+ undergraduates; developed supplemental materials improving exam performance by 15%.

KEY PROJECTS

Analytical Characterization & Stability
Chemistry of Materials 2025

Connected molecular structure, purity, and degradation behavior using orthogonal spectroscopy, electrochemistry, and thermal methods.

FT-IR, UV-Vis-NIR, CV, TGA/DSC, NMR, GPC, MS

AFOSR Thin-Film Process Development
DoD Funded

Established process-property relationships for CVD/surface-preparation workflows and reproducible conditions.

CVD, QCM-D, surface prep, thin films

Bio-Based Polymer Scale-Up Controls
Current

Defined DOE-guided process windows and material-quality specifications for reproducible polymer scale-up.

DOE, SPC, polymer characterization

EDUCATION

Ph.D. in Chemistry - University of Colorado Boulder 2025

Dissertation: Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials
Advisor: Prof. Seth Marder
(Transferred from Georgia Institute of Technology)

B.S. (Research), Major: Chemistry; Minor: Materials Science - Indian Institute of Science (IISc) 2019

INSPIRE Scholarship for Higher Education (Top 1% nationally)

CERTIFICATIONS

Statistical Process Control (SPC) - Coursera, 2026

TECHNICAL SKILLS

Analytical Characterization: LC-MS, GC-MS, NMR, GPC/SEC, FT-IR, UV-Vis-NIR, TGA, DSC, QCM-D, cyclic voltammetry
Materials & Stability: Organic/small-molecule materials, polymers, thin films, solid-state/thermal analysis, degradation/failure analysis
Process & Documentation: DOE, SPC, JMP (basic), Python (basic), Excel, SOPs, technical reports, reproducibility studies

JOURNAL PUBLICATIONS

1. Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Narrow-Band Electrochromism of a Ferrocene-Substituted Rhodamine B." Chemistry of Materials, 2025, 37, 5558-5568.
2. Mandal, J.; Ramasamy Selvaraju, V.; Yan, W.; Divandari, M.; Spencer, N. D.; Dubner, M. "In Situ Monitoring of SI-ATRP Throughout Multiple Reinitiations Under Flow by Means of a Quartz Crystal Microbalance." RSC Advances, 2018, 8, 20048-20055.
3. Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Beyond Rhodamine B: Investigating Alternative Xanthene Cores for Modular Electrochromic Systems." (Manuscript under preparation).

CONFERENCE PRESENTATIONS

Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials, ACS Fall National Meeting, Denver, CO, August 2024.
Electrochemical and Spectroscopic Properties of Substituted Rhodamine Derivatives, Gerischer-Lewerenz Electrochemistry Symposium, Fort Collins, CO, August 2024.
Decoupling Redox and Color for High-Contrast Electrochromics, Excited State Processes Conference, Santa Fe, NM, June 2023.
Electrochromism: Decoupling Color and Redox in Rhodamine-Based Systems, Rocky Mountain Solid State Chemistry Workshop, Boulder, CO, January 2023.

HONORS & AWARDS

Plastic Sustainability Innovation in Material Award | INSPIRE Scholarship (Top 1%, India, 2015) | 97.12 Percentile - IIT JEE Mains (2015) | All India Rank 144 - NEST (2014)